Notes: Sirri and Tufano (JF, 1998)

Costly Search and Mutual Fund Flows

Contents

1 Big picture

- **Question.** How do households decide which mutual funds to buy, and how much of fund inflows can be explained by past performance versus costly search and marketing?
- **Setting.** The paper studies U.S. open-end equity mutual funds from **1971–1990**, using a large fund-level panel with data on returns, flows, fees, risk, fund-family size, and media coverage.
- **Core challenge.** A positive relation between past fund returns and subsequent inflows could reflect rational updating about manager skill, but it could also reflect salience, marketing, brand recognition, broker incentives, or media attention. Likewise, weak outflows from poor performers could be due to survivorship bias or to investors simply not noticing bad news.
- **Main conceptual contribution.** The paper brings *costly search* into the mutual-fund flow literature. Investors do not choose among perfectly visible, commodity-like products. Instead, they are more likely to buy funds that are easier to notice and easier to remember.
- **Main hypotheses.**
  - In a costless-search world, flows should respond positively to good performance, negatively to bad performance, negatively to risk, and negatively to fees.
  - With costly search, investors should disproportionately buy funds that are easy to identify, such as those in large fund complexes, those marketed more heavily, and those receiving more media attention.
  - If good performance is a salient characteristic, then the performance-flow relation should be strongest where search costs are lowest or marketing effort is highest.
- **Main empirical design.** The authors estimate annual fund-level flow regressions with lagged performance, risk, fees, fund size, and objective-category flows, reporting **Fama–MacBeth** averages across yearly cross-sections. They then interact performance with proxies for search costs: fund-family size, fee-based marketing intensity, and media attention.
- **Main baseline result.** Fund flows respond *very strongly* to top performance, but there is little or no penalty for poor performance. In the baseline piecewise specification, the top-quintile performance slope is about **1.63–1.69**, while the bottom-quintile slope is near **zero**.
- **Key pricing result.** Higher fees are associated with lower flows. A **100 basis point** difference in fees is associated with about a **2.9 percentage point** difference in annual flows.
- **Key search-cost result.** Funds in larger complexes grow faster, and high-fee funds—interpreted as funds with stronger marketing and sales effort—have a much steeper top-performance-flow relation. In Table V, the top-performance slope is about **1.12** for low-fee funds versus about **2.25** for high-fee funds.
- **Key asymmetry result.** The central asymmetry is that investors flock to recent winners but do not flee recent losers at the same rate.
- **Main credibility result.** The asymmetry survives alternative return measures, different return horizons, a check for survivorship bias using "dead" funds, fee-change specifications, and tests separating load from no-load funds.

- **Main economic takeaway.** Mutual fund inflows are shaped not only by performance, but by the visibility of performance. Search costs, marketing, and salience help explain why investor money chases winners.

2 Data and measurement (key objects)

2.1 Why the mutual fund industry is the right laboratory

- Mutual funds are a clean setting for studying household financial choice because they package portfolio decisions into a discrete product that consumers must actively select.
- The paper argues that selecting a fund resembles choosing a large consumer durable more than solving a full-time portfolio problem: households face many options, rely on advertising and intermediaries, and do not continuously process all available information.
- This framing motivates the paper's emphasis on search costs, brand awareness, and salience.

Interpretation. The point is not just that returns matter. It is that returns matter in an environment where investors do not observe or process all funds symmetrically.

2.2 Main data source and sample construction

- The main source is the **Investment Company Data Institute (ICDI)** database.
- The sample covers open-end U.S. equity mutual funds from **December 1971 through December 1990**.
- The paper keeps the three major equity objectives:
  - aggressive growth,
  - growth and income,
  - long-term growth.
- Specialty sector funds and international funds are excluded.
- The full sample contains **690 funds** offered by **288 fund families / complexes**.

The ICDI data provide:

- monthly net asset values (NAVs),
- distributions,
- quarterly total net assets (TNA),
- and classification variables such as age, objective, and distribution method.

The dataset is supplemented by hand-collected fee information from *Wiesenberger Investment Reports*, *Barron's*, and *Morningstar Mutual Fund Values*.

2.3 Coverage and survivorship

- Over the full 20-year sample, the main dataset covers about **87%** of total equity-fund assets and about **71%** of the number of funds.
- The data are biased toward funds still alive in 1990, because ICDI dropped dead funds from its historical database.
- Fortunately, the authors obtained separate information on **58** funds that died during **1987–1990**, which gives them an unbiased subperiod sample for a direct survivorship-bias test.

Interpretation. For most of the paper the sample is weighted toward surviving, larger funds, but the authors explicitly test whether this matters for the core asymmetric flow result.

2.4 What Table I says about the industry

Table I describes a rapidly growing industry:

- Number of funds rises from **228** in 1971 to **632** in 1990.
- Total equity-fund assets rise from **$35.5 billion** to **$171.7 billion**.
- Mean complex assets rise from about **$329 million** to **$588 million**.
- The mean number of equity funds per complex rises from **2.53** to **8.28**.
- The share of funds charging a load falls from **74%** to **52%**.
- Mean expense ratios rise from **0.96%** to **1.44%**.
- Mean annual total fees (expense ratio plus amortized load) move from **1.66%** to **1.37%**.

Interpretation. The mutual fund sector grew dramatically, fund families became much larger, and pricing structure shifted away from front-end loads toward ongoing expenses.

2.5 Defining net fund flows

The core dependent variable is the growth of fund assets net of reinvested returns:

$$\mathrm{FLOW}_{i,t} = \frac{\mathrm{TNA}_{i,t} - \mathrm{TNA}_{i,t-1}(1 + R_{i,t})}{\mathrm{TNA}_{i,t-1}}. \tag{1}$$

where:

- $\mathrm{TNA}_{i,t}$ is total net assets of fund i at time t,
- $R_{i,t}$ is the fund's return over period t.

Interpretation. This is the percentage growth in assets that cannot be explained by capital gains or reinvested distributions. It is intended to capture net new money flowing into or out of the fund.

2.6 Performance measures

The paper emphasizes performance measures that were visible to actual investors in the 1970s and 1980s.

The main baseline measure is a **fractional performance rank** within objective category and year. For each objective-year, a fund is assigned a rank from **0** (worst) to **1** (best) based on its prior raw return.

The paper also studies:

- three-year and five-year raw return rankings,
- lagged one-year rankings from years $t-1$, $t-2$, and $t-3$,
- market-model excess returns,
- Jensen one-factor alphas estimated using prior monthly data.

Interpretation. The paper deliberately starts with simple measures because those are the measures retail investors were most likely to observe.

2.7 Risk measures

The baseline risk measure is the standard deviation of monthly fund returns over the prior year. The paper also checks longer windows, such as over months -1 to -24 and -1 to -36.

Interpretation. The goal is not to recover the "correct" asset-pricing risk measure; the goal is to proxy for the risk information actually salient to retail fund buyers.

2.8 Fee measures

Total fees are defined as:

$$\text{Total Fee} = \text{Expense Ratio} + \frac{1}{7} \times \text{Load}, \tag{2}$$

where the front-end load is amortized over a **seven-year** holding period, the average holding period for an equity mutual fund in the data.

The paper separately examines:

- the level of total fees,
- changes in total fees,
- changes in loads,
- changes in expense ratios.

Interpretation. A fee can matter both as a *price* paid by the investor and as a *proxy for marketing effort*, especially when the fee finances loads or 12b-1 style distribution activities.

2.9 Search-cost proxies

The paper uses three practical proxies for search costs:

- **Fund-family / complex size:** log of total net assets managed by the fund family.
- **Marketing and distribution effort:** proxied by total fees, especially loads and distribution-related charges.
- **Media attention:** each fund's share of circulation-weighted citations in nine financial periodicals and eleven major newspapers.

Interpretation. These are imperfect proxies, but all try to capture how easy it is for a retail investor to notice and recall a fund.

3 Models and empirical specifications (all equations + interpretation)

3.1 Costless-search benchmark and hypotheses

The paper first states predictions from a benchmark world in which information is free:

- bad past performance should reduce flows if poor performance persists,
- good past performance should raise flows if winning may persist,
- higher risk should reduce flows,
- higher fees should reduce flows.

Interpretation. These are the predictions against which the actual data are compared. The later costly-search analysis asks why the empirical response is so asymmetric.

3.2 Baseline multivariate flow regression

The general specification is written in the paper as

$$\mathrm{FLOW}_{i,t} = f(\mathrm{Return}_{i,t-1}, \mathrm{Risk}_{i,t-1}, \mathrm{Fees}_{i,t-1}, \mathrm{OBJFLOW}_t, \log \mathrm{TNA}_{i,t-1}). \tag{3}$$

In practice, this becomes a linear fund-level regression with controls for:

- lagged fund size $\log(\mathrm{TNA}_{i,t-1})$,
- growth of the fund's objective category,
- lagged return-based performance,
- lagged return volatility,
- fee levels and, later, fee changes.

Interpretation. Objective-category flow captures broad shifts into or out of a fund style. Fund size captures the mechanical fact that a given dollar inflow implies a larger percentage growth rate for small funds.

3.3 Piecewise linear performance specification

To capture asymmetry, performance ranks are decomposed into segments. In the main three-part version:

$$\text{LOWPERF}_{i,t-1} = \min(\text{RANK}_{i,t-1}, 0.2), \tag{4}$$
$$\text{MIDPERF}_{i,t-1} = \min\big(0.6, \text{RANK}_{i,t-1} - \text{LOWPERF}_{i,t-1}\big), \tag{5}$$
$$\text{HIGHPERF}_{i,t-1} = \text{RANK}_{i,t-1} - (\text{LOWPERF}_{i,t-1} + \text{MIDPERF}_{i,t-1}). \tag{6}$$

The estimating equation is therefore of the form

$$\begin{aligned}\text{FLOW}_{i,t} = \alpha + \beta_1\text{LOWPERF}_{i,t-1} + \beta_2\text{MIDPERF}_{i,t-1} + \beta_3\text{HIGHPERF}_{i,t-1} \\ + \beta_4\text{Risk}_{i,t-1} + \beta_5\text{Fees}_{i,t-1} + \beta_6\text{OBJFLOW}_t + \beta_7 \log\text{TNA}_{i,t-1} + u_{i,t}.\end{aligned} \tag{7}$$

Interpretation. Instead of forcing a single linear slope, the paper lets the response to performance differ for losers, middle performers, and winners.

3.4 Estimation method

- The authors estimate regressions *year by year* and then average the coefficients across years using the **Fama–MacBeth (1973)** procedure.
- This is more conservative than pooling all fund-years and helps account for serial dependence within funds over time.

Interpretation. The key reported coefficients are time-series means of annual cross-sectional slopes, not a single pooled OLS estimate.

3.5 Baseline coefficient pattern: Table II

Table II contains the core empirical pattern.

In column (A), using five performance quintiles:

- bottom quintile slope: **-0.007** (essentially zero),
- fourth quintile: **0.104**,
- third quintile: **0.283**,
- second quintile: **0.061**,
- top quintile: **1.693**.

In the more compact column (B):

- **LOWPERF = -0.035** (insignificant),
- **MIDPERF = 0.170** (significant),
- **HIGHPERF = 1.633** (highly significant).

Other baseline coefficients are also intuitive:

- log lag TNA: about **-0.048**, so smaller funds grow faster in percentage terms;
- category flow: about **0.95**, so fund inflows comove strongly with category-wide inflows;
- volatility: about **-1.04** to **-1.07**, marginally significant;
- total fees: about **-0.029**, marginally significant.

Interpretation. Investors aggressively chase winners. The asymmetry is economically large: moving five percentiles within the top quintile is associated with about an **8.4 percentage point** larger flow, versus roughly **0 to 1.4 percentage points** in other ranges.

3.6 Alternative performance and risk specifications: Table III

Table III checks robustness using different return horizons and more sophisticated risk-adjusted measures.

- Three-year and five-year raw return rankings still produce the same broad asymmetry: top performers get large inflows.
- When performance from years $t-1$, $t-2$, and $t-3$ are entered separately, the most recent performance matters the most, though earlier performance still has some predictive power.
- Rankings based on market-model excess returns and Jensen's alpha also generate a strong winner-chasing pattern.
- When raw-return ranks and alpha are included together, raw return ranks still matter, suggesting that simple reported performance has its own salience beyond formal risk adjustment.

Interpretation. Households seem to react to the kind of performance measure that is easy to see and easy to market, not just to academically cleaned-up alpha.

3.7 Fee-change specifications: Table IV

Table IV adds fee changes to the basic model.

- Change in total fees enters negatively: about **-0.119**.
- When separated, fee *increases* have little effect, but fee *decreases* matter strongly: the coefficient on fee decreases is about **-0.213**.
- Decomposing fees further, changes in **expense ratios** matter, but changes in **loads** do not.

- In column (C), the coefficient on change in expenses is about **-0.253**; change in load is about **0.004** and insignificant.

Interpretation. Investors react much more when ongoing expenses fall than when front-end load schedules change. This fits the paper's view that loads partly finance selling effort, so their price effect and search-cost effect may offset each other.

3.8 Search-cost interaction specifications: Table V

The core costly-search test interacts performance with search-cost proxies.

Complex size.

- The coefficient on log complex TNA is about **0.029** to **0.034**.
- The paper notes that, around the 1980 mean complex size of **$316 million**, an extra **$100 million** under management predicts slightly less than a **1 percentage point** increase in next-year flows.
- But the interaction of large-complex status with performance is near zero and insignificant.

High-fee / high-marketing funds.

- In column (B), the top-performance slope for low-fee funds is **1.123**.
- The interaction **HIGHPERF** $\times$ **HIGHFEE** is **1.127**, implying a top-performance slope of roughly **2.25** for high-fee funds.
- In column (C), after allowing for both complex-size and high-fee interactions, the top-performance interaction with HIGHFEE remains large at about **1.230**.

Interpretation. Larger complexes seem easier to find in general, but they do not appear more performance-sensitive. By contrast, high-fee funds—interpreted as better marketed funds—turn good performance into inflows much more effectively.

3.9 Media coverage construction and equations

The media variable is built from Lexis/Nexis mentions in **20 major publications** (9 financial periodicals and 11 newspapers), weighted by circulation.

For each year and objective category, the paper constructs a fund's *media share*: its share of circulation-weighted citations among funds in that objective.

The paper then studies both:

- the determinants of media share,
- and whether media share predicts flows.

3.10 Determinants of media share: Table VI

In Table VI, the dependent variable is each fund's share of circulation-weighted media citations.

Main coefficients:

- own size (log lag TNA): about **0.003**, strongly positive;
- total fees: about **0.001**, marginally positive;
- return volatility: about **0.027**, marginally positive;
- complex size: about **-0.001**, slightly negative conditional on own size;
- LOWPERF: **-0.019**;
- HIGHPERF: **0.028**.

Interpretation. Media attention is roughly *U-shaped* in performance: both very good and very bad funds get covered, while middle performers get less attention. This is important because it means the asymmetric winner-chasing result cannot simply be blamed on journalists only talking about winners.

3.11 Media attention and fund flows: Table VII

Table VII relates flows to media attention.

- Current media share is strongly positively associated with current growth: coefficient about **3.055**.
- But lagged media share (about **0.997**) and lagged residual media share (about **1.360**) are not statistically significant.
- Interaction terms between media-attention dummies and performance are mostly insignificant.

Interpretation. Current media attention and flows move together, but the paper is cautious: this could be reverse causality. Once lagged or residualized media is used, the evidence that media causally amplifies performance sensitivity becomes weak.

4 Identification and empirical design (what makes the estimates credible)

4.1 What identifies the baseline estimate

The key identifying variation comes from differences across funds in lagged relative performance, fees, and risk within the same broad objective category and year, after controlling for category-wide flows and lagged fund size.

Interpretation. The paper is not using a quasi-experiment. Identification comes from cross-sectional fund heterogeneity plus careful lag structure and robustness checks.

4.2 Why reverse causality is limited in the baseline flow tests

- The main explanatory variables are lagged by one year.
- Current inflows cannot mechanically cause prior-year performance.
- The strongest baseline pattern therefore runs from past outcomes to future money growth.

Caveat. This does not rule out omitted variables that jointly predict both prior performance and future flows, but it greatly weakens simple simultaneity concerns.

4.3 Survivorship bias

A natural concern is that the missing dead funds mechanically weaken the observed penalty for poor performance. The authors address this directly:

- they use the unbiased 1987–1990 sample that includes dead funds,
- assign a **-100%** flow in the year a fund dies,
- and reestimate the base model.

The asymmetry remains: strong winner-chasing but little response to poor performance.

Interpretation. The weak loser effect is not just a survivorship artifact.

4.4 Common shocks and style flows

The regressions control for **objective-category flows**. This is important because funds in the same investment style may all receive inflows in a bull market or outflows in a downturn.

Interpretation. The coefficient on past fund performance is therefore not just picking up the fact that all growth funds had a good year together.

4.5 Why the paper's search-cost proxies are suggestive but imperfect

- **Complex size** can proxy for lower search costs, but it can also proxy for broader services or easier within-family switching.
- **Fees** measure price, but they also partly measure marketing and distribution effort.
- **Media share** captures attention, but the paper cannot classify stories as positive or negative.

Interpretation. The paper's identification of search costs is based on patterns across several imperfect proxies that fit together, rather than on one clean instrument.

4.6 Adverse selection between load and no-load investors

One concern is that the asymmetry might come from the type of investor who buys broker-sold load funds. Perhaps these investors are less likely to withdraw from bad funds. The paper tests this by estimating the models separately for load and no-load funds and finds the same broad asymmetry in both groups.

Interpretation. The results are not driven only by broker-dominated funds.

4.7 Media endogeneity

The strongest media result is contemporaneous, which is hard to interpret causally. The authors therefore use:

- lagged media share,
- lagged residual media share from the Table VI media-determinants regression.

Neither produces strong evidence of a robust lagged media effect on flows.

Interpretation. The paper is careful not to oversell media coverage as a causal amplifier.

4.8 What the paper can and cannot distinguish

The paper can credibly show that:

- flows chase top performance much more than they punish bad performance,
- fees and fee cuts matter,
- marketing-related proxies shape how strongly performance translates into flows.

But it cannot fully separate:

- rational return-chasing from behavioral extrapolation,
- pure search costs from service-quality or family-level convenience,
- media attention from other correlated forms of visibility,
- price effects of fees from all search-cost effects of fees.

4.9 Relation to the literature

- Relative to **Gruber (1996)** and **Zheng (1998)**, this paper focuses less on whether fund flows are "smart money" and more on how households actually choose funds.
- Relative to **Brown et al. (1996)** and **Chevalier and Ellison (1995)**, it highlights how convex flow-performance effects create option-like incentives for managers.
- Relative to consumer-choice and marketing work such as **Capon, Fitzsimons, and Prince (1996)**, it brings brand awareness and costly search directly into a finance setting.

5 Tables and figures

5.1 Table I: cross-sectional characteristics of the mutual fund sample

Read.

- The industry grows dramatically from 1971 to 1990 in number of funds, total assets, and family breadth.
- Load funds become less dominant, while expense ratios rise.
- By 1990, the mean family offers many more funds than in 1971.

Economic message. The environment investors face becomes richer and more crowded over time, which makes search and brand recognition increasingly relevant.

Table I

Cross-Sectional Characteristics of the Equity Mutual Fund Sample in 1971, 1980, and 1990

The sample includes open-end U.S. funds that have an investment objective of aggressive growth, growth and income, or long-term growth, as classified by the Investment Company Data Institute. "Number of funds" reflects the number of equity funds in our sample still in existence as of the end of the year. The total number of different funds in our sample is 690, but by the end of 1990, 58 of these funds ceased to exist. "Fund complexes" are separate families of mutual funds which must sell at least one equity fund to be included in our sample. "Total fee" is estimated as expense ratio plus amortized load, where the load is amortized without discounting over seven years, which is the average holding period for an equity fund in these data. For complex and individual fund data, the mean and standard deviation (S.D.) are reported for each characteristic.

		1971	1980	1990
Equity fund sector characteristics				
Number of funds		228	264	632
Total fund assets ($billions)		$35.5	$36.7	$171.7
Fund complex characteristics				
Assets managed ($millions)	Mean	$329.4	$315.6	$ 588.2
	S.D.	$669.2	$593.0	$2269.7
Number of funds sold by complex	Mean	2.53	3.39	8.28
	S.D.	1.96	3.03	12.89
Individual equity fund characteristics				
Fund assets ($millions)	Mean	$185	$176	$272
	S.D.	$383	$306	$785
Charging load		74%	53%	52%
Load charged (for load funds)	Mean	8.29%	7.80%	5.25%
	S.D.	1.07%	1.34%	1.51%
Annual expense ratios	Mean	0.96%	1.05%	1.44%
	S.D.	0.67%	0.56%	0.82%
Total fee (annual)	Mean	1.66%	1.47%	1.37%
	S.D.	6.37%	7.40%	9.51%

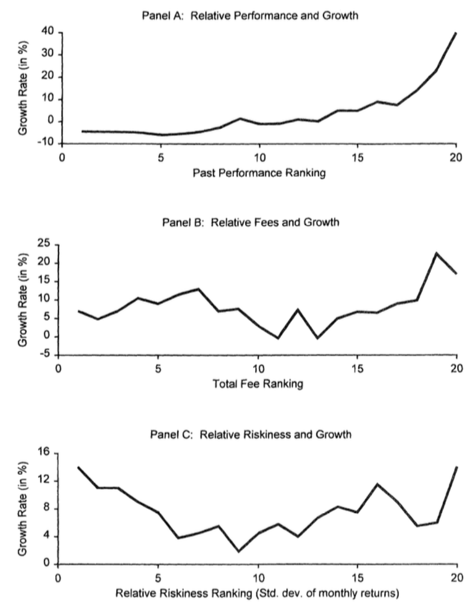

Figure 1. The mean growth rate of 690 open-end U.S. equity mutual funds as a function of their relative performance, fee levels, and return volatility, 1970 through 1990. For each year from 1971 to 1990, funds are ordered within one of three objective categories (aggressive growth, growth and income, and long-term growth), and divided into 20 equal groups based on their total return, level of fees, and portfolio return volatility, respectively. For each of these 20 groups, the mean growth rate of the funds in that group is calculated. The growth rate is defined as $(\mathrm{TNA}_t - \mathrm{TNA}_{t-1}) * (1 + \mathrm{R}_t) / \mathrm{TNA}_{t-1}$, where TNA_t is the total net assets of the fund at time t and R_t is the return of the fund in period t. The top panel divides the samples into groups based on the prior years' total return. The middle panel divides the sample into groups based on the prior years' level of total fees (which equals the expense ratio plus one-seventh of the load). The bottom panel divides the sample into groups based on the prior year's standard deviation of monthly returns, a measure of return volatility.

5.2 Figure 1: performance, fees, risk, and growth

Read.

- Panel A shows a sharp hockey-stick relation between past return rank and next-year growth.

Table II

The Effect of Relative Performance, Return Volatility, and Fee Levels on the Growth of 690 Equity Mutual Funds, 1971 through 1990

The sample includes open-end U.S. funds that have an investment objective of aggressive growth, growth and income, or long-term growth, as classified by the Investment Company Data Institute. The table reports OLS coefficient estimates using the growth rate of net new money as the dependent variable, which is defined as $(\mathrm{TNA}_{i,t} - \mathrm{TNA}_{i,t-1}) * (1 + R_{i,t})/(\mathrm{TNA}_{i,t-1})$, where $\mathrm{TNA}_{i,t}$ is fund i's total net assets at time t, and $\mathrm{R}_{i,t}$ is the raw return of fund i in period t. The independent variables include the log of fund i's total net assets in the prior period (Log lag $\mathrm{TNA}_{i,t-1}$), the growth rate of net new money for all funds in the same investment category as fund i (Flows to fund category), the volatility of the prior year's monthly returns, the level of total fees (expense ratio plus amortized load) charged by the fund for an investor with a seven-year holding period, and measures of the fractional performance rank of fund i in the preceding years. A fund's fractional rank (RANK_t) represents its percentile performance relative to other funds with the same investment objective in the same period, and ranges from 0 to 1. In this table, fractional ranks are defined on the basis of a funds' one-year raw return. The coefficients on fractional ranks are estimated using a piecewise linear regression framework over five quintiles in column (A). For example, the 5th or bottom performance quintile (LOWPERF) is defined as $\mathrm{Min}(\mathrm{RANK}_{t-1}, 0.2)$, the 4th performance quintile is defined as $\mathrm{Min}(0.2, \mathrm{RANK}_{t-1} - \mathrm{LOWPERF}_{t-1})$, and so forth, up to the highest performance quintile (HIGHPERF). In column (B), the middle three performance quintiles are combined into one grouping labeled as the 2nd–4th performance quintile (MIDPERF), defined as Min (0.6, RANK − LOWPERF). The coefficients on these piecewise decompositions of fractional ranks represent the slope of the performance-growth relationship over their range of sensitivity. These regressions are run year-by-year, and standard errors and t-statistics are calculated from the vector of annual results, as in Fama and MacBeth (1973). p-values are given in parentheses below the coefficient estimates.

Independent Variable	(A)	(B)
Intercept	0.245	0.252
	(0.002)	(0.001)
Log lag TNA	−0.048	−0.048
	(0.000)	(0.000)
Flows to fund category	0.949	0.965
	(0.004)	(0.003)
Std. dev. of monthly returns	−1.043	−1.068
	(0.105)	(0.101)
Total fees	−0.029	−0.029
	(0.061)	(0.057)
Breakdown of RANK		
Bottom performance quintile (LOWPERF)	−0.007	−0.035
	(0.971)	(0.843)
4th performance quintile	0.104	—
	(0.355)	
3rd performance quintile	0.283	—
	(0.009)	
2nd–4th performance quintiles (MIDPERF)	—	0.170
		(0.000)
2nd performance quintile	0.061	—
	(0.694)	
Top performance quintile (HIGHPERF)	1.693	1.633
	(0.000)	(0.000)
Adjusted R^2	14.2%	14.3%
Number of observations	4098	4098

- The bottom 80% of funds exhibit only a shallow slope; the top 20% get a dramatic growth bonus.
- Panel B shows no simple monotone fee-growth relation in the raw data.
- Panel C shows no simple monotone risk-growth relation either.

Economic message. The raw data already suggest that the main action is in *winner chasing*; fees and risk need multivariate analysis because their univariate relations are confounded.

5.3 Table II: the baseline performance-flow regression

Read.

- The top-performance coefficient is huge and highly significant.
- The bottom-performance coefficient is near zero.
- Higher fees and higher volatility reduce flows; smaller funds grow faster in percentage terms.

Economic message. The central empirical fact is asymmetric: investors strongly reward recent winners and barely punish losers.

5.4 Table III: alternative performance and risk measures

Table III

The Effect of Alternative Performance and Risk Measures on the Growth of Equity Mutual Funds, 1971 through 1990

The sample includes open-end U.S. funds that have an investment objective of aggressive growth, growth and income, or long-term growth, as classified by the Investment Company Data Institute. The table reports OLS coefficient estimates for seven separate regressions using the growth rate of net new money as the dependent variable, which is defined as $(\mathrm{TNA}_{i,t} - \mathrm{TNA}_{i,t-1}) * (1 + \mathrm{R}_{i,t})/(\mathrm{TNA}_{i,t-1})$, where $\mathrm{TNA}_{i,t}$ is fund i's total net assets at time t, and $\mathrm{R}_{i,t}$ is the raw return of fund i in period t. The independent variables used in all seven regressions include the log of fund i's total net assets in the prior period (Log lag $\mathrm{TNA}_{i,t-1}$) and the growth rate of net new money for all funds in the same investment category as fund i (Flows to fund category). The regressions also include various measures of the fractional performance rank (RANK_t) of fund i in the preceding years. The performance ranks are divided into three unequal groupings. For example, the bottom performance grouping (LOWPERF) is the lowest quintile of performance, defined as Min(RANK_{t-1}, 0.2). The middle three performance quintiles are combined into one grouping (MIDPERF) defined as Min (0.6, RANK − LOWPERF), and the highest performance quintile (HIGHPERF) is defined as RANK − (LOWPERF + MIDPERF). The coefficients on these piecewise decompositions of fractional ranks represent the slope of the performance-growth relationship over their range of sensitivity. These fractional ranks are defined using the prior three-year raw returns (column A), the prior five-year raw returns (column B), and each of the prior three-years' raw returns separately (column C). Column D uses a measure of RANK based on market-model excess returns, defined as $\mathrm{R}_{i,t} - \mathrm{E}(\mathrm{R}_{i,t})$, where $\mathrm{E}(\mathrm{R}_{i,t})$ is defined using an estimate of beta from past monthly returns. Columns E and F report regressions using a measure of RANK based on estimates of Jensen's alpha over the preceding five years. Column G uses the fractional ranks from the prior one-year raw return and the level of the five-year Jensen's alpha. Because the raw return measures in columns A through C do not incorporate any risk adjustment, we also include the volatility of monthly returns over the same period in which the performance is measured. These regressions are run year-by-year, and standard errors and t-statistics are calculated from the vector of annual results, as in Fama and MacBeth (1973). p-values are given in parentheses below the coefficient estimates.

	(A)	(B)	(C)	(D)	(E)	(F)	(G)
Intercept	0.241 (0.025)	0.211 (0.107)	0.165 (0.103)	0.191 (0.002)	0.143 (0.041)	0.092 (0.134)	0.204 (0.006)
Log lag of TNA	−0.043 (0.000)	−0.033 (0.001)	−0.042 (0.000)	−0.049 (0.000)	−0.033 (0.001)	−0.031 (0.000)	−0.027 (0.001)
Flows to fund category	0.758 (0.049)	0.831 (0.051)	0.737 (0.049)	0.898 (0.014)	0.971 (0.013)	0.988 (0.011)	0.883 (0.008)
Standard deviation of monthly returns							
Over months −1 to −36	−1.007 (0.343)	—	—	—	—	—	—
Over months −1 to −60	—	−0.792 (0.511)	—	—	—	—	—

	(A)	(B)	(C)	(D)	(E)	(F)	(G)
Standard deviation of monthly returns							
Over months −1 to −12	—	—	−0.053 (0.950)	—	—	—	—
Over months −13 to −24	—	—	−0.955 (0.243)	—	—	—	—
Over months −25 to −36	—	—	−1.047 (0.150)	—	—	—	—
Percentile ranking based on raw returns							
Bottom quintile, year −1 (LOWPERF)	—	—	0.030 (0.902)	—	—	0.136 (0.547)	−0.204 (0.121)
Middle three quintiles, year −1 (MIDPERF)	—	—	0.183 (0.001)	—	—	0.107 (0.041)	0.092 (0.054)
Top quintile, year −1 (HIGHPERF)	—	—	1.027 (0.012)	—	—	0.899 (0.045)	1.164 (0.016)
Bottom quintile, year −2 (LOWPERF)	—	—	0.055 (0.781)	—	—	—	—
Middle three quintiles, year −2(MIDPERF)	—	—	0.124 (0.027)	—	—	—	—
Top quintile, year −2 (HIGHPERF)	—	—	0.313 (0.108)	—	—	—	—
Bottom quintile, year −3(LOWPERF)	—	—	−0.097 (0.679)	—	—	—	—
Middle three quintiles, year −3 (MIDPERF)	—	—	0.083 (0.156)	—	—	—	—
Top quintile, year −3 (HIGHPERF)	—	—	0.401 (0.015)	—	—	—	—
Bottom quintile, years −1 to −3 (LOWPERF)	−0.047 (0.769)	—	—	—	—	—	—
Middle three quintiles, years −1 to −3 (MIDPERF)	0.215 (0.006)	—	—	—	—	—	—

Table III—*Continued*

	(A)	(B)	(C)	(D)	(E)	(F)	(G)
Percentile ranking based on raw returns							
Top quintile, years −1 to −3 (HIGHPERF)	0.998 (0.000)	—	—	—	—	—	—
Bottom quintile, years −1 to −5 (LOWPERF)	—	−0.176 (0.435)	—	—	—	—	—
Middle three quintiles, years −1 to −5 (MIDPERF)	—	0.174 (0.003)	—	—	—	—	—
Top quintile, years −1 to −5 (HIGHPERF)	—	1.545 (0.000)	—	—	—	—	—
Percentile ranking based on excess returns							
Bottom quintile, year −1 (LOWPERF)	—	—	—	0.189 (0.128)	—	—	—
Middle three quintiles, year −1 (MIDPERF)	—	—	—	0.145 (0.002)	—	—	—
Top quintile, year −1 (HIGHPERF)	—	—	—	1.465 (0.001)	—	—	—
Jensen's alpha (5 years)	—	—	—	—	—	—	19.851 (0.000)
Percentile ranking based on Jensen's alpha							
Bottom quintile, years −1 to −5 (LOWPERF)	—	—	—	—	−0.157 (0.275)	−0.212 (0.200)	—
Middle three quintiles, years −1 to −5 (MIDPERF)	—	—	—	—	0.285 (0.003)	0.194 (0.011)	—
Top quintile, years −1 to −5 (HIGHPERF)	—	—	—	—	2.165 (0.000)	1.842 (0.000)	—
Adjusted R^2	14.0%	16.6%	12.9%	12.9%	16.9%	20.6%	10.5%
Number of observations	3286	2675	3286	4098	2675	2675	2675

Read.

- The winner-chasing pattern survives three-year and five-year raw returns.
- It also survives market-model excess returns and Jensen's alpha.
- Recent performance is more powerful than older performance when several lags are included together.

Economic message. The asymmetry is not an artifact of one arbitrary return measure.

5.5 Table IV: fee changes

Table IV

The Effect of Fee Changes on the Growth of Equity Mutual Funds, 1971 through 1990

The sample includes open-end U.S. funds that have an investment objective of aggressive growth, growth and income, or long-term growth, as classified by the Investment Company Data Institute. The table reports OLS coefficient estimates for four separate regressions using the growth rate of net new money as the dependent variable, which is defined as $(\mathrm{TNA}_{i,t} - \mathrm{TNA}_{i,t-1}) * (1 + \mathrm{R}_{i,t})/(\mathrm{TNA}_{i,t-1})$, where $\mathrm{TNA}_{i,t}$ is fund i's total net assets at time t, and $\mathrm{R}_{i,t}$ is the raw return of fund i in period t. The independent variables used in all four regressions include the log of fund i's total net assets in the prior period (Log lag $\mathrm{TNA}_{i,t-1}$), the log of the total net assets of all funds in fund i's complex in the prior period (Log lag complex $\mathrm{TNA}_{i,t-1}$), the growth rate of net new money for all funds in the same investment category as fund i (Flows to fund category), and the standard deviation of monthly returns. The new independent variables in this table are the change in total fees (column A), which is broken down into fee increases and decreases (column B), and changes in expense ratios and loads (column C), broken down into expense and load increases and decreases (column D). The regressions also include measures of the fractional performance rank (RANK_t) of fund i based on raw returns in the preceding year. The performance ranks are divided into three unequal groupings. The bottom performance grouping (LOWPERF) is the lowest quintile of performance, defined as Min(RANK_{t-1}, 0.2). The middle three performance quintiles are combined into one grouping (MIDPERF) defined as Min (0.6, RANK − LOWPERF), and the highest performance quintile (HIGHPERF) is defined as RANK − (LOWPERF + MIDPERF). The coefficients on these piecewise decompositions of fractional ranks represent the slope of the performance-growth relationship over their range of sensitivity. These regressions are run year-by-year, and standard errors and t-statistics are calculated from the vector of annual results, as in Fama and MacBeth (1973). p-values are given in parentheses below the coefficient estimates.

	(A)	(B)	(C)	(D)
Intercept	0.124 (0.020)	0.121 (0.035)	0.114 (0.039)	0.104 (0.080)
Log lag TNA	−0.079 (0.000)	−0.077 (0.000)	−0.080 (0.000)	−0.075 (0.000)
Log lag complex TNA	0.035 (0.002)	0.035 (0.002)	0.036 (0.001)	0.037 (0.001)
Flows to fund category	1.172 (0.002)	1.161 (0.003)	1.156 (0.004)	1.108 (0.008)
Std. dev. of monthly returns	−0.820 (0.213)	−0.947 (0.181)	−0.688 (0.279)	−0.988 (0.191)
Total fees	−0.026 (0.126)	−0.034 (0.056)	−0.026 (0.089)	−0.050 (0.008)
Change in total fees	−0.119 (0.015)	—	—	—
Increase in total fees	—	0.042 (0.529)	—	—
Decrease in total fees	—	−0.213 (0.012)	—	—
Change in expenses	—	—	−0.253 (0.001)	—
Increase in expenses	—	—		0.070 (0.126)
Decrease in expenses	—	—	—	−0.689 (0.000)

Table IV—*Continued*

	(A)	(B)	(C)	(D)
Change in load	—	—	0.004 (0.442)	—
Increase in load	—	—	—	0.022 (0.614)
Decrease in load	—	—	—	0.011 (0.106)
Bottom performance quintile (LOWPERF)	0.109 (0.430)	0.105 (0.454)	0.111 (0.404)	0.173 (0.240)
Middle three performance quintiles (MIDPERF)	0.161 (0.001)	0.159 (0.001)	0.155 (0.001)	0.151 (0.000)
Top performance quintile (HIGHPERF)	1.697 (0.001)	1.692 (0.001)	1.638 (0.001)	1.530 (0.001)
Adjusted R^2	19%	20%	20%	19%
Number of observations	3721	3721	3721	3721

Read.

- Reductions in fees, especially expense ratios, are associated with higher inflows.
- Load changes matter much less than expense-ratio changes.
- The basic performance asymmetry remains intact.

Economic message. Investors respond to changes in the cost of holding a fund, but the relevant price seems more tied to ongoing expenses than to sales commissions.

5.6 Table V: complex size, marketing, and the cost of search

Table V

The Impact on the Growth of Equity Mutual Funds of Variables that Capture the Extent to Which Consumers Face Search Costs in Identifying Fund Investments, 1971 through 1990

The sample includes open-end U.S. funds that have an investment objective of aggressive growth, growth and income, or long-term growth, as classified by the Investment Company Data Institute. The table reports OLS coefficient estimates for separate regressions using the growth rate of net new money as the dependent variable, which is defined as $(\mathrm{TNA}_{i,t} - \mathrm{TNA}_{i,t-1}) * (1 + \mathrm{R}_{i,t}) / (\mathrm{TNA}_{i,t-1})$, where $\mathrm{TNA}_{i,t}$ is fund i's total net assets at time t, and $\mathrm{R}_{i,t}$ is the raw return of fund i in period t. The independent variables used in the regressions include the log of fund i's total net assets in the prior period (Log lag $\mathrm{TNA}_{i,t-1}$), the log of the total net assets of all funds in fund i's complex in the prior period (Log lag complex $\mathrm{TNA}_{i,t-1}$), the growth rate of net new money for all funds in the same investment category as fund i (Flows to fund category), the standard deviation of monthly returns, the total fees charged (expense ratio plus amortized load), and changes in total fees. The regressions include measures of the fractional performance rank (RANK_t) of fund i based on raw returns in the preceding year. The performance ranks are divided into three unequal groupings. The bottom performance grouping (LOWPERF) is the lowest quintile of performance, defined as $\mathrm{Min}(\mathrm{RANK}_{t-1}, 0.2)$. The middle three performance quintiles are combined into one grouping (MIDPERF) defined as Min (0.6, RANK − LOWPERF), and the highest performance quintile (HIGHPERF) is defined as RANK − (LOWPERF + MIDPERF). The coefficients on these piecewise decompositions of fractional ranks represent the slope of the performance-growth relationship over their range of sensitivity. The regression in Column A includes an interaction term that is the product of a dummy that equals one only for complexes whose assets under management are larger than the median complex for year $t-1$ (LARGE) times the fund's performance ranking. The regression in Column B includes an interaction term that is the product of a dummy that equals one only for funds whose fees are above the median for their objective category for year $t-1$ (HIGHFEE) times the fund's performance ranking. Column C includes both sets of these interaction terms. These regressions are run year-by-year, and standard errors and t-statistics are calculated from the vector of annual results, as in Fama and MacBeth (1973). p-values are given in parentheses below the coefficient estimates.

	(A)	(B)	(C)
Intercept	0.167	0.121	0.138
	(0.010)	(0.053)	(0.040)
Log lag TNA	−0.077	−0.077	−0.077
	(0.000)	(0.000)	(0.000)
Flows to fund category	1.142	1.074	1.092
	(0.002)	(0.005)	(0.004)
Std. dev. of monthly returns	−0.954	−1.187	−1.172
	(0.123)	(0.082)	(0.072)
Total fees	−0.027	−0.012	−0.015
	(0.099)	(0.530)	(0.474)
Change in total fees	−0.098	−0.112	−0.100
	(0.028)	(0.011)	(0.015)
Log lag complex TNA	0.029	0.034	0.031
	(0.029)	(0.001)	(0.024)
Bottom performance quintile (LOWPERF)	0.054	0.081	0.085
	(0.731)	(0.669)	(0.635)
Middle three performance quintiles (MIDPERF)	0.128	0.215	0.205
	(0.011)	(0.004)	(0.005)
Top performance quintile (HIGHPERF)	1.646	1.123	1.007
	(0.005)	(0.003)	(0.066)

Table V—*Continued*

	(A)	(B)	(C)
Interaction terms:			
Large-complex dummy times...			
Bottom performance quintile	0.040	—	0.009
(LOWPERF * LARGE)	(0.834)		(0.964)
Middle three performance quintiles	0.041	—	0.033
(MIDPERF * LARGE)	(0.577)		(0.666)
Top performance quintile	0.046	—	0.212
(HIGHPERF * LARGE)	(0.939)		(0.761)
Interaction terms:			
High-total-fee dummy times...			
Bottom performance quintile	—	−0.069	−0.034
(LOWPERF * HIGHLFEE)		(0.579)	(0.788)
Middle three performance quintiles	—	−0.117	−0.132
(MIDPERF * HIGHFEE)		(0.143)	(0.089)
Top performance quintile	—	1.127	1.230
(HIGHPERF * HIGHFEE)		(0.029)	(0.048)
Adjusted R^2	18.8%	18.7%	20.1%
Number of observations	3858	3858	3858

Read.

- Funds in larger complexes grow faster on average.
- But large-complex funds do *not* exhibit a stronger performance-flow slope.
- High-fee funds exhibit much stronger winner-chasing than low-fee funds.

Economic message. Brand visibility helps flows generally, but marketing intensity appears especially important in turning strong performance into strong inflows.

5.7 Table VI: which funds get media attention?

Read.

- Bigger funds, higher-fee funds, and more volatile funds get more media attention.
- Media attention is U-shaped in performance: both very good and very bad funds get covered.

Economic message. Journalists do not only cover winners, so the paper's winner-chasing result is not simply a byproduct of bad funds receiving no coverage at all.

Table VI

The Determinants of the Share of Circulation-Weighted Media Citations Received by Equity Mutual Funds, 1971 through 1990

The sample includes open-end U.S. funds that have an investment objective of aggressive growth, growth and income, or long-term growth, as classified by the Investment Company Data Institute. The dependent variable in this OLS analysis is each fund's media share, defined as the share of circulation-weighted media citations received by each fund as a percentage of the total circulation-weighted media citations received by other funds with the same objective in each year. This measure is constructed from the number of stories about each fund appearing in Lexis-Nexis over the period 1971 to 1990 in twenty leading financial periodicals and newspapers, weighting each cite by that periodical's circulation. The independent variables used in the regressions include the log of fund i's total net assets in the prior period (Log lag $TNA_{i,t-1}$), the log of the total net assets of all funds in fund i's complex in the prior period (Log lag complex $TNA_{i,t-1}$), the standard deviation of monthly returns, and the total fees charged (expense ratio plus amortized load). The regression includes the fractional performance rank ($RANK_t$) of fund i based on raw returns in the preceding year. The performance ranks are divided into three unequal groupings. The bottom performance grouping (LOWPERF) is the lowest quintile of performance, defined as Min($RANK_{t-1}$, 0.2). The middle three performance quintiles are combined into one grouping (MIDPERF) defined as Min (0.6, RANK − LOWPERF), and the highest performance quintile (HIGHPERF) is defined as RANK − (LOWPERF + MIDPERF). The coefficients on these piecewise decompositions of fractional ranks represent the slope of the performance-growth relationship over their range of sensitivity. p-values are given in parentheses below the coefficient estimates.

Independent Variable	Coefficient (p-value)
Intercept	0.001
	(0.534)
Log lag TNA	0.003
	(0.000)
Total fees	0.001
	(0.061)
Std. dev. of monthly returns	0.027
	(0.066)
Log lag complex TNA	−0.001
	(0.000)
Bottom performance quintile (LOWPERF)	−0.019
	(0.044)
Middle three performance quintiles (MIDPERF)	0.002
	(0.327)
Top performance quintile (HIGHPERF)	0.028
	(0.000)
Adjusted R^2	4.39%
Number of observations	3305

Table VII

The Impact of the Amount of Media Attention Received by Funds on Their Growth, 1971 through 1990

The sample includes open-end U.S. funds that have an investment objective of aggressive growth, growth and income, or long-term growth, as classified by the Investment Company Data Institute. The table reports OLS coefficient estimates for separate regressions using the growth rate of net new money as the dependent variable, which is defined as $(TNA_{i,t} - TNA_{i,t-1}) * (1 + R_{i,t})/(TNA_{i,t-1})$, where $TNA_{i,t}$ is fund i's total net assets at time t, and $R_{i,t}$ is the raw return of fund i in period t. The independent variables used in the regressions include the log of fund i's total net assets in the prior period (Log lag $TNA_{i,t-1}$), the log of the total net assets of all funds in fund i's complex in the prior period (Log lag complex $TNA_{i,t-1}$), the growth rate of net new money for all funds in the same investment category as fund i (Flows to fund category), the standard deviation of monthly returns, the total fees charged (expense ratio plus amortized load), and changes in total fees. The regressions include measures of the fractional performance rank ($RANK_t$) of fund i based on raw returns in the preceding year. The performance ranks are divided into three unequal groupings. The bottom performance grouping (LOWPERF) is the lowest quintile of performance, defined as Min($RANK_{t-1}$, 0.2). The middle three performance quintiles are combined into one grouping (MIDPERF) defined as Min (0.6, RANK − LOWPERF), and the highest performance quintile (HIGHPERF) is defined as RANK − (LOWPERF + MIDPERF). The coefficients on these piecewise decompositions of fractional ranks represent the slope of the performance-growth relationship over their range of sensitivity. The additional independent variables in this table attempt to capture the extent of search costs faced by consumers. In column A, we add each fund's media share (Share of media by objective), defined as the share of circulation-weighted media citations received by each fund as a percentage of the total circulation-weighted media citations received by other funds with the same objective in each year, and interaction terms equal to the product of the three performance groupings with an indicator variable (CURMED) that equals one only if the fund is in the 90th percentile of media attention that year for its objective category. In column B, we add each fund's lagged media share (Lagged share of media by objective) and interaction terms equal to the product of the three performance groupings with an indicator variable (LAGMED) that equals one only if the fund is in the 90th percentile of media attention for the previous year for its objective category. In column C, we add an independent variable defined using the regression specified in Table VI as the residual of the regression of media attention on a group of independent variables (Lagged residual share of media), and interaction terms equal to the product of the performance groupings with an indicator variable (RESMED) that equals one only if the fund has a media regression residual in the 90th percentile of all media residuals for the previous year for its objective category. These regressions are run year-by-year, and standard errors and t-statistics are calculated from the vector of annual results, as in Fama and MacBeth (1973). p-values are given in parentheses below the coefficient estimates.

	(A)	(B)	(C)
Intercept	0.104	0.257	0.041
	(0.196)	(0.228)	(0.507)
Log lag TNA	−0.102	−0.080	−0.053
	(0.000)	(0.000)	(0.000)
Log lag complex TNA	0.044	0.069	0.027
	(0.004)	(0.065)	(0.000)
Flows to fund category	0.383	4.943	0.979
	(0.632)	(0.214)	(0.011)
Std. dev. of returns	−0.599	−1.058	−0.476
	(0.321)	(0.223)	(0.419)
Total fees	−0.007	−0.146	−0.017
	(0.722)	(0.267)	(0.253)
Change in total fees	0.048	0.056	−0.049
	(0.725)	(0.665)	(0.152)

Table VII—*Continued*

	(A)	(B)	(C)
Bottom performance quintile (LOWPERF)	−0.038	5.101	0.202
	(0.919)	(0.333)	(0.108)
Middle three performance quintiles (MIDPERF)	0.136	−1.771	0.116
	(0.144)	(0.365)	(0.027)
Top performance quintile (HIGHPERF)	1.509	1.314	0.951
	(0.009)	(0.117)	(0.002)
Level of media attention			
Share of media by objective	3.055	—	—
	(0.001)		
Lagged share of media by objective	—	0.997	—
		(0.262)	
Lagged residual of share of media	—	—	1.360
			(0.262)
Interaction terms: Media times performance			
Current media above median times			
Bottom quartile performance (LOWPERF * CURMED)	0.143	—	—
	(0.634)		
Middle three quartiles performance (MIDPERF * CURMED)	0.008	—	—
	(0.952)		
Top quartile performance (HIGHPERF * CURMED)	0.341	—	—
	(0.738)		
Prior year media above median times			
Bottom quartile performance (LOWPERF * LAGMED)	—	−4.850	—
		(0.352)	
Middle three quartiles performance (MIDPERF * LAGMED)	—	1.977	—
		(0.332)	
Top quartile performance (HIGHPERF * LAGMED)	—	0.165	—
		(0.882)	
Prior year residual media above median times			
Bottom quartile performance (LOWPERF * RESMED)	—	—	−0.087
			(0.375)
Middle three quartiles performance (MIDPERF * RESMED)	—	—	0.092
			(0.211)
Top quartile performance (HIGHPERF * RESMED)	—	—	0.479
			(0.300)
Adjusted R^2	18.2%	24.1%	20.0%
Number of observations	3129	2993	2825

5.8 Table VII: media attention and fund growth

Read.

- Current media attention is positively associated with current growth.
- But lagged and residualized media measures are weak predictors of future flows.
- Interactions between media and performance are not robust.

Economic message. Media matter descriptively, but the paper's stronger search-cost evidence comes from marketing intensity rather than from lagged press coverage.

6 Short summary

- Mutual fund investors chase recent winners much more than they punish recent losers.
- Higher fees reduce flows overall, but high-fee funds benefit more from strong performance, consistent with the role of marketing and salience.
- Funds in large complexes enjoy higher baseline inflows, but not stronger winner-chasing.

- Media attention is correlated with contemporaneous growth, but lagged-media evidence is weak.
- The paper's main contribution is to show that costly search helps explain why performance-flow relations are so asymmetric.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that mutual fund flows are not explained by a simple symmetric response to risk-adjusted performance. Instead, fund inflows are strongly nonlinear: money floods into recent winners, but recent losers do not experience matching outflows. This pattern survives multiple alternative specifications and does not appear to be driven by survivorship bias.

7.2 Economic intuition

Investors do not process the mutual fund universe like professional portfolio managers. They face search costs, depend on salient information, and are influenced by marketing and visibility. Strong recent performance becomes a particularly effective sales attribute when it is easy to communicate and easy to notice.

7.3 Takeaway

The mutual fund industry provides a microfoundation for return-chasing behavior. Performance matters, but the *visibility* of performance matters too. Winner-chasing is therefore partly a product of market outcomes and partly a product of how those outcomes are marketed, distributed, and socially recognized.